

RECOMMENDER SYSTEMS · INDIVIDUAL PROJECT

A Movie Recommender Prototype

From non-personalized baselines to matrix factorization on MovieLens 32M

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Objective & approach

One prototype, many algorithms, the same evaluation protocol

1

Build

Every technique becomes a working, reusable module in a single codebase.

2

Compare

All models share one dataset, split and metric set — results are directly comparable.

3

Think critically

Accuracy is not enough: we also measure coverage, novelty and popularity bias.

Track chosen: Movies — MovieLens 32M (genres as content metadata).

Dataset & exploratory analysis

MovieLens 32M · half-star ratings 0.5–5 · genre metadata

200,948

users

84,432

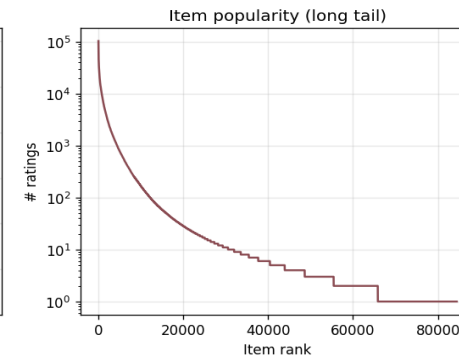
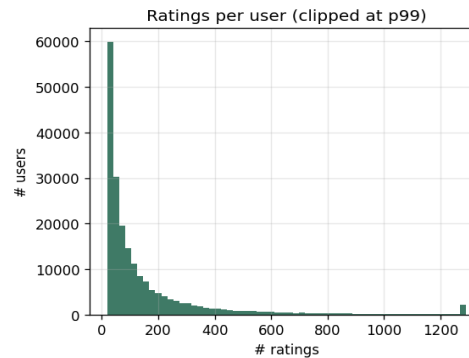
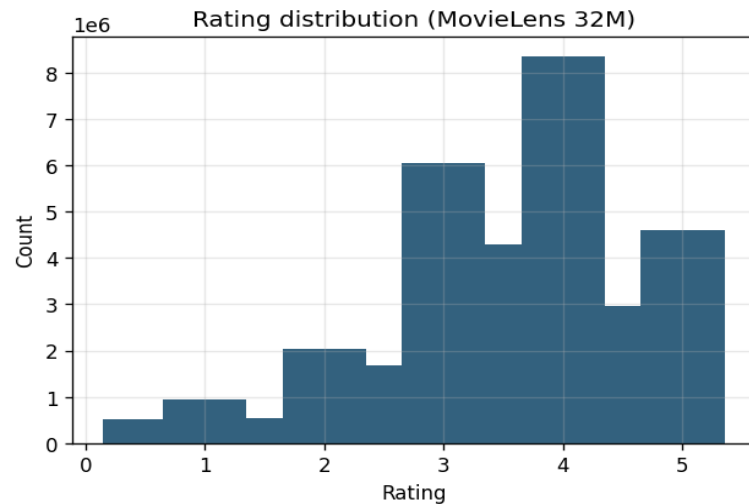
rated movies

32.0M

ratings

99.8%

sparsity



What the data tells us

- Ratings skew positive (mean ≈ 3.5)
- Strong long tail: few blockbusters, many niche films
- 99.8% sparse $\rightarrow$ cold cells everywhere
- Popularity bias will shape results

Preprocessing & evaluation protocol

Same pipeline for every model — fair comparison

Preprocessing

- Cap catalog to 3,000 most-popular items + k-core (≥ 100 / ≥ 20 ratings)
- Sub-sample 10,000 users $\rightarrow$ dense, fast, responsive prototype
- Working set: ~ 1.34 M ratings, cached to disk (float32 $\rightarrow$ fits free cloud)
- Per-user TEMPORAL 80/20 split $\rightarrow$ no train-on-future leakage

Metrics

- Accuracy: Precision@10, Recall@10, Hit-Rate@10, NDCG@10, MRR
- Relevance = held-out rating ≥ 4.0
- Beyond-accuracy: catalog coverage & novelty (self-information)
- Rating prediction: MAE / RMSE (item-item CF); k & factors tuned
- Averaged over a fixed user sample, identical for all models

Algorithms implemented

Nine models in code, from simple baselines to latent factors

● Most Popular

Rank items by interaction count.
Non-personalized baseline.

● Highest / Bayesian avg

Hard min-ratings cutoff vs. IMDb
damped mean (course formula).

● Random

Lower-bound sanity check / coverage
ceiling.

● Content-based

TF-IDF & raw-genre vectors; mean-
centered profile; cosine.

● Item-Item CF

Adjusted-cosine item similarity, top-
k=10 neighbourhood.

● User-User CF

Mean-centered user similarity, top-
k=40 neighbours.

● Matrix Factorization

Truncated SVD (sklearn): $R \approx U \cdot V^T$, dot-
product scores, argsort.

All share one interface

`fit()` then `recommend(user, n)`.
Methods are hot-swappable in the
pipeline and the UI.

Results — accuracy vs. beyond-accuracy

Top-10 · temporal split (leakage-free) · 10k users · 95% CI on NDCG

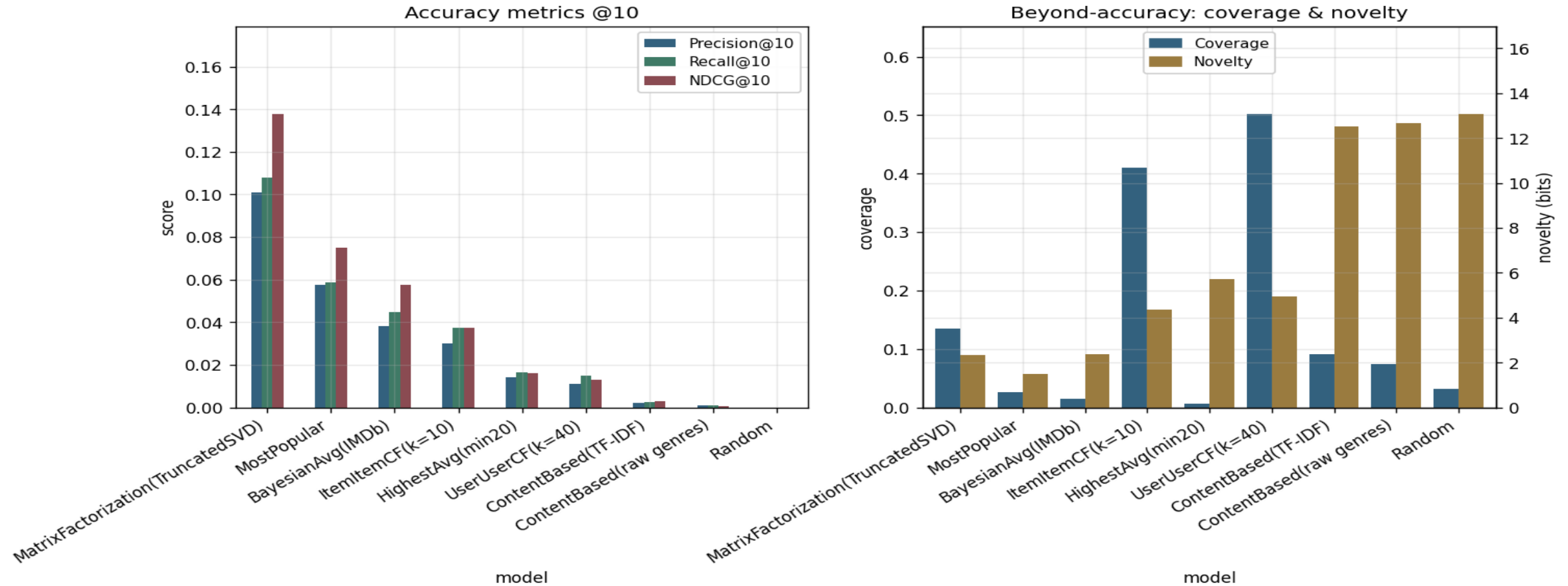
Model	Prec@10	Rec@10	NDCG@10	±CI	Cov	Novelty
Matrix Factorization (SVD)	0.101	0.108	0.138	0.022	0.135	2.35
Most Popular	0.058	0.059	0.075	0.014	0.027	1.51
Bayesian Average (IMDb)	0.038	0.045	0.058	0.013	0.015	2.38
Item-Item CF (k=10)	0.030	0.037	0.037	0.008	0.410	4.36
Highest Average (min 20)	0.014	0.016	0.016	0.005	0.006	5.72
User-User CF (k=40)	0.011	0.015	0.013	0.005	0.502	4.96
Content-based (TF-IDF)	0.002	0.003	0.003	0.003	0.091	12.51
Content-based (raw)	0.001	0.001	0.001	0.001	0.074	12.66
Random	0.000	0.000	0.000	—	0.032	13.07

Read it this way

- SVD leads significantly — its CI does not overlap popularity's
- Temporal split: no train/test leakage
- Bayesian (IMDb) avg beats the hard-threshold average
- CF covers 41–50% of the catalog; content-based is the most novel

The accuracy-discovery trade-off

Same models, two very different rankings



Recommendation examples

Same user (id 124204), top-5 by method — the personality of each model

● Most Popular

Jurassic Park · Raiders of the Lost Ark · The Usual Suspects · The Dark Knight · Inception

● Bayesian Avg (IMDb)

The Usual Suspects · Dr. Strangelove · Spirited Away · 12 Angry Men · Casablanca

● Content-based (genres)

King Lear · I Fidanzati · Il Posto · Deepwater Horizon · Queen of Katwe

● Item-Item CF (k=10)

Planet Earth II · Lawrence of Arabia · Capturing the Friedmans · The King's Speech

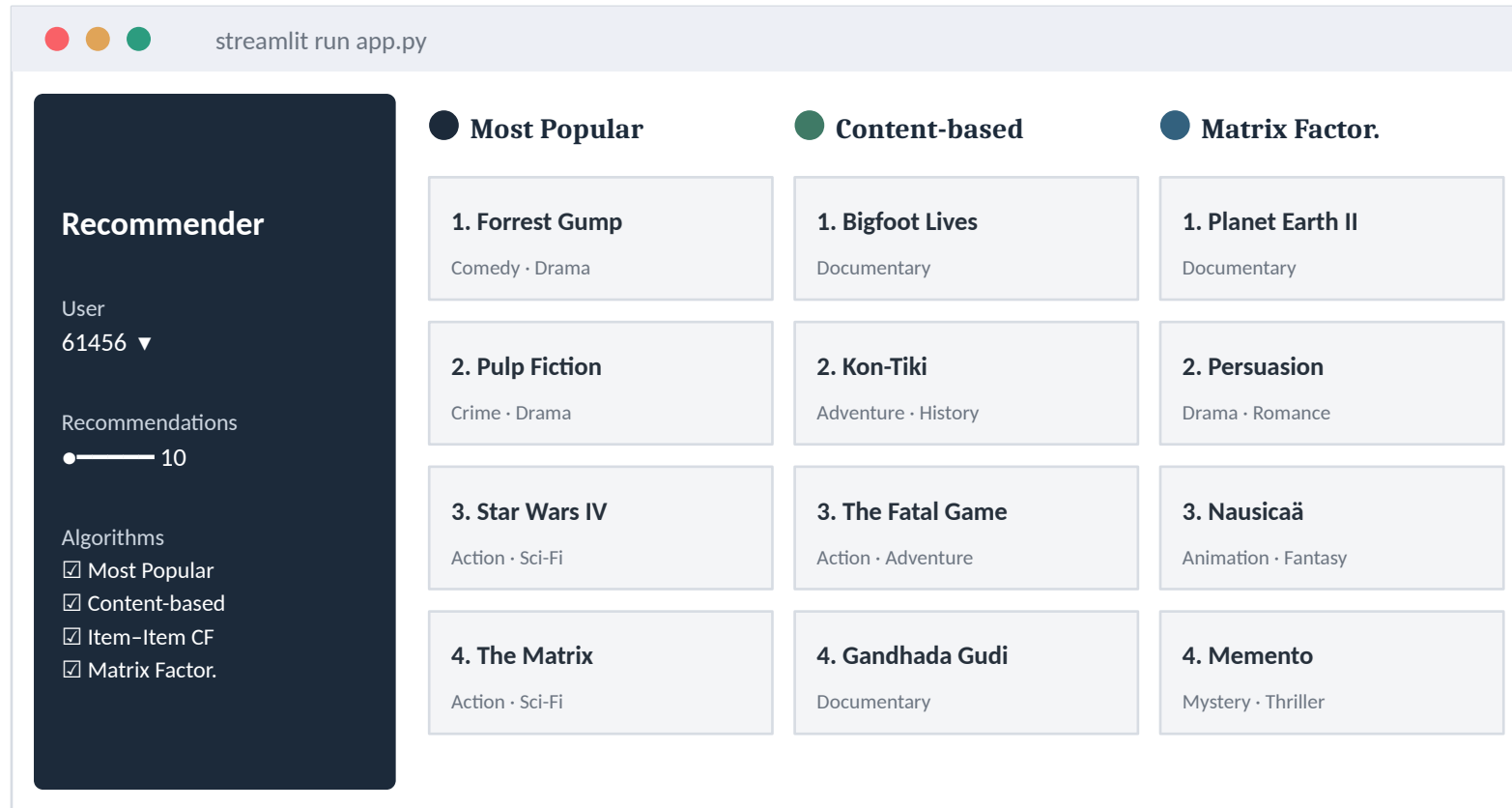
● Matrix Factorization (SVD)

Memento · The Usual Suspects · Raiders of the Lost Ark · The Green Mile · Monsters, Inc.

Popularity / Bayesian = mainstream & acclaimed · Content = genre-coherent but obscure · CF & SVD = strongly personalized picks.

Interactive prototype (UX)

Streamlit app — onboard a new user or explore an existing one



What the demo offers

● New-user questionnaire

Cold-start onboarding: a few questions build a taste profile and recommend films — no ratings needed.

● Side-by-side comparison

See how each algorithm recommends for the same user — the trade-off made tangible.

● Taste profile + "more like this"

Top-rated movies, favourite-genre chart, and content-based similar-movie search.

● Live & cached

Models train once; switching user, method or answers is instant.

Technical challenges

What was hard, and how it was solved

Scale & speed

32M ratings (877 MB) made naive CF/MF infeasible. Solved with compact dtypes, item-cap + k-core filtering, user sub-sampling and vectorized, cached NumPy scoring.

Course-aligned MF

Followed the course method: sklearn TruncatedSVD on the sparse matrix, dot-product scores, argsort top-N — no fragile native dependencies.

Fair evaluation

Cold users break personalized models. A per-user split guarantees overlap; identical user sample & relevance rule for all models.

Beyond accuracy

Accuracy alone hides popularity bias. Added coverage & novelty to expose the trade-off.

Strengthening the evaluation

Methodological issues identified during review, and how each was addressed

Random split leaked the future

✓ Temporal split → NDCG $\approx 0.33 \rightarrow \approx 0.14$ (realistic)

A single point estimate

✓ 95% confidence intervals → SVD's lead is significant

Popularity bias was hidden

✓ Long-tail analysis → SVD $\sim 1\%$ tail vs 49% item-CF

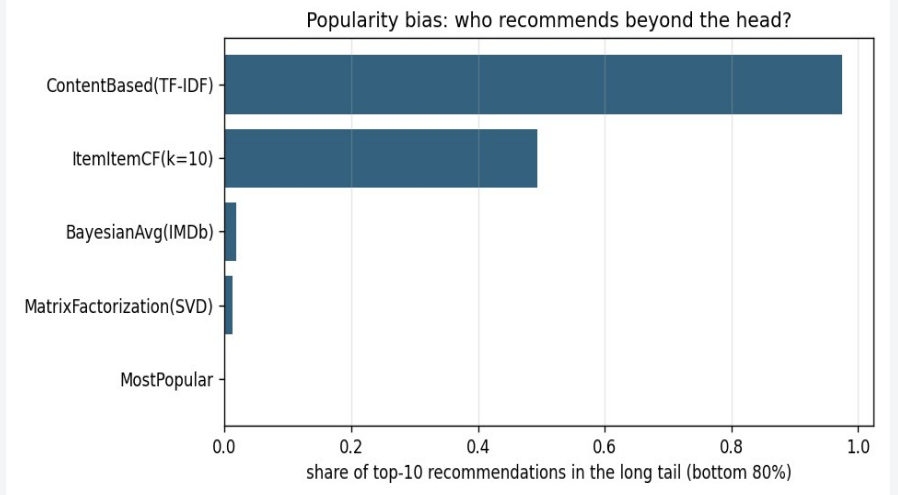
No support for brand-new users

✓ Questionnaire onboarding → cold-start solved in the app

SVD imputes missing ratings as 0

✓ Documented as a known explicit-feedback trade-off

Who recommends beyond the popular head?



Long-tail share of each model's top-10. The SVD "winner" stays on the head; CF and content-based reach the tail.

Threats to validity & next steps

What the numbers do — and don't — show

Threats to validity

- Leakage: fixed with a temporal split (SVD NDCG $\approx 0.33 \rightarrow \approx 0.14$)
- Significance: 95% CIs reported — top ranking is real, not noise
- Popularity bias: SVD puts only ~1% of recs in the long tail (vs 49% item-CF)
- Dense popular subset $\rightarrow$ optimistic; not vs. published benchmarks
- SVD imputes missing = 0; no true cold-start eval

Next steps

- Richer content features (tags, cast, synopsis), not just genres
- Hybrid: blend SVD accuracy with content/CF novelty
- Explicit diversity & popularity-debiasing in the ranking
- A proper cold-start (new user / new item) evaluation
- Deploy the Streamlit demo online (already built — see UX slide)

Conclusions

1. **No single best model.**

The right choice depends on the objective — engagement, discovery or coverage.

2. **Accuracy is not the whole story.**

The most accurate model (popularity / SVD) is also the most popularity-biased.

3. **Personalization adds value.**

Matrix factorization and CF surface relevant items a popularity list would miss.

Thank you. Live demo: movierec-fp.streamlit.app